14 Olivia and Jessica have in total half as many sweets as Fran and Gary have in total.

Fran and Gary share their sweets in the ratio 2 : 3
Olivia and Jessica share their sweets in the ratio 9 : 1

Fran got w sweets.
Gary got x sweets.
Olivia got y sweets.
Jessica got z sweets.

Find, in its simplest form, $w : x : y : z$

Fran : Gary = 2 : 3, total 5 parts

Olivia : Jessica = 9 : 1, total 10 parts

Their total is half the Fran/Gary total.

Scale the first ratio by 4: 8 : 12

Then the combined ratio is 8 : 12 : 9 : 1

1 mark
group totals 5 and 10

1 mark
half-total scale factor

1 mark
8 : 12 with 9 : 1

1 mark
8 : 12 : 9 : 1

8 : 12 : 9 : 1

(Total for Question 14 is 4 marks)

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